Varun Srivastava

408-896-0580 | varun.neal@berkeley.edu | linkedin.com/in/varun-n-sri/ | varunneal.github.io

Education

University of California, Berkeley — Graduated Dec 2023
B.A. Computer Science, Mathematics — GPA: 3.9/4

Experience

Machine Learning Engineer II — August 2024 – Present
AdMarketplace — New York, NY

- Leading training for the vector search model powering retrieval for our full database of products (> 100 million titles), driving major increase in ad relevance and revenue.
- Designing ML time-series models for the firm's algorithmic ad bidding solution, optimizing millions of dollars in daily ad impressions.
- Built a full-stack prototype for an AI shopping assistant; implemented an end-to-end RAG pipeline with embedding-based retrieval and LLM generation.

Founding Engineer — August 2022 – December 2022
Stork Oracle Network — San Francisco, CA

- Designed a high-speed and reliable WebSocket API for live crypto data. The pricefeed served several notable paying DeFi clients, such as Vertex and Apex.
- Scaled the full-stack Python infrastructure up to meet high uptime and low latency requirements for platforms trading millions of dollars daily.

Quantitative Analysis Intern — May 2022 – August 2022
Dexterity Capital — Seattle, WA

- Implemented CMA-ES strategy in forecasting models, improving hyperparameter convergence by > 100x.
- Analyzed signals in cryptocurrency markets that became incorporated in the company's proprietary high frequency trading (HFT) market making algorithm, which trades billions of dollars in volume a day.

Projects

Modded NanoGPT | *https://github.com/KellerJordan/modded-nanogpt*

- Achieved many world records in a popular LLM training competition (4000 Github stars), with several records involving novel pretraining algorithmic research.

Spotify MCP | *github.com/varunneal/spotify-mcp*

- MCP Server integrating Spotify with LLMs (> 500 GitHub stars). Winner at the Anthropic MCP Hackathon.

Semantic Finder | *do-me.github.io/SemanticFinder*

- Developed a semantic search site powered by LLMs and a corresponding Chrome extension (> 300 GitHub stars).

Research & Publications

Machine Learning & Vision Researcher | *Berkeley Visual Computing Lab* — December 2023 – May 2024

- Lee, J., **Srivastava, V.**, Jennings, N., Ng, R. "Theory of Human Tetrachromatic Color Experience." *ACM SIGGRAPH*, 2024. Honorable Mention for Technical Papers Awards, spotlighted at the annual conference.
- Researched the emergence of color vision in humans via neurological models, specifically in the case of esoteric conditions such as human tetrachromacy.

Vision & Neuroscience Researcher | *Na-Ji Biophysics Lab* — September 2019 – April 2020

- Implemented computer vision tools in Matlab for researchers to track the pupil size of rats in video data.
- Correlated terabytes of time-series data with pupil measurements to analyze how rodents react to shock stimulus.

Technical Skills

Languages: Python (PyTorch, Triton, Pandas), CUDA, C/C++, JavaScript, Go, Java, SQL, Matlab.
Frameworks & development: AWS, Databricks, Docker, Kubernetes Node, React, Django.